LIST OF PUBLICATIONS

Peer-reviewed publications from PhD thesis.

1. **Thesis Chapter-5: Singha S**, Ericsson CD, Chowdhury S, Nath SC, Paul OB, Hoque MA, Boqvist S, Persson Y, Rahman MM. *Occurrence and etiology of subclinical mastitis in water buffalo in Bangladesh*. Journal of Dairy Research. 2021;88(3):314-320. doi: 10.1017/S0022029921000698. PMID: 34412714 **(Published** on 20.08.2021**)**.
2. **Thesis Chapter-6: Singha S**, Koop G, Ceciliani F, Derks M, Hoque MA, Hossain MK, Howlader MMR, Rahman MM, Khatun M, Boqvist S, Persson Y. *The prevalence and risk factors of subclinical mastitis in water buffalo (Bubalis bubalis) in Bangladesh*. Research in Veterinary Science. 2023;158:17-25. doi: 10.1016/j.rvsc.2023.03.004. PMID: 36907020 **(Published** on 08.03.2023**)**.
3. **Thesis Chapter-7: Singha S**, Koop G, Rahman MM, Ceciliani F, Addis MF, Howlader MMR, Hossain MK, Piccinini R, Locatelli C, Persson Y, Bronzo V. *Pathogen group-specific risk factors for intramammary infection in water buffalo* **(Revisions received** in **Plos One** on 18.12.2023**).**
4. **Thesis Chapter-8: Singha S.**, Ceciliani F, Rahman MM, Mannan MA, Chowdhury S, Nath, SC, Paul OB, Persson Y, Boqvist S. *Factors influencing somatic cell counts and bacterial contamination in unpasteurized milk obtained from water buffalo in Bangladesh*. Tropical animal health and production. 2023; 55(4), 242. https://doi.org/10.1007/s11250-023-03644-x. PMID: 37328619 **(Published** on 16.06.2023**)**.
5. **Thesis Chapter-9: Singha S.**, Koop G, Rahman MM, Ceciliani F, Howlader MMR, Boqvist S, Cremonesi P, Hoque MN, Persson Y, Lecchi C. *Foodborne pathogens in milk and milk products along the water buffalo milk chain in Bangladesh* **(Submitted** in **Scientific Reports** on 21.11.2023**)**

WATER BUFFALO MILK CHAIN IN BANGLADESH

S SINGHA

ABSTRACT

The present study aimed to assess udder health by estimating the prevalence of subclinical mastitis (SCM) and identifying intramammary infection (IMI) causing pathogens. Furthermore, milk quality was evaluated by measuring bulk milk somatic cell count (BMSCC) on buffalo farms and total bacterial count (TBC), total non-aureus staphylococci count (TNAS), total *Staphylococcus* (*S.*) *aureus* count (TSA), total Enterobacteriaceae count (TEC), and the presence of zoonotic pathogens along the water buffalo milk chain in Bangladesh. A total of 248 water buffalo farms from nine buffalo concentration districts, comprising 51 free-range, 106 semi-free-range, 33 households, 54 semi-intensive, and four intensive buffalo farms, were visited to collect individual quarter milk to diagnose SCM using CMT; BMSCC was also measured on-farms. One hundred twenty-two farm bulk milk samples, 109 middlemen milk samples, 111 milk collection centers, and 35 milk product samples were collected and assessed for TBC, TNAS, TSA, and TEC. A subset of these samples, comprising farm bulk milk (n = 34), the middleman (n = 37), the milk collection center (n = 37), and milk products from shops (n = 35), were analyzed using real-time PCR to estimate the prevalence of seven important zoonotic pathogens: *S. aureus*, *Escherichia* (*E.*) *coli*, Shiga toxin-producing *E. coli* O157:H7, *Campylobacter* (*C.*) *jejuni*, *Listeria* (*L.*) *monocytogenes*, *Salmonella* (*S.*) *enterica*, and *Yersinia* (*Y.*) *enterocolitica*. Quarter, buffalo, farm level, and milk value chain-related data was collected using questionnaires and observations. The overall SCM prevalence was high at the quarter level (27.9 %) and the buffalo level (51.5 %). Non-aureus staphylococci were the predominant IMI-associated bacterial species. The proportion of non-aureus staphylococci or *Mammaliicoccus sciuri* (NASM), *S. aureus*, and other bacterial species identified in the buffalo quarter samples varied between buffalo farms. An intensive buffalo-rearing system, teats on the left, funnel-shaped teats, udder asymmetry, single milker, and a lack of quarantine facility were all associated with poor buffalo udder health. The geometric mean of BMSCC was 217,000 cells per mL of milk, which was low on average, but some farms could improve substantially. Farm-hygiene-related risk factors for IMI were poor milking hygiene and cleanliness of the hind quarters. The likelihood of IMI from any pathogen and NASM were higher in buffalo herds with poor milking hygiene. Poor cleanliness of the hind quarters increased the chances of IMI caused by any pathogen, NASM, and *S. aureus*. Farm management-related risk factors for IMI were the milking frequency and buffalo source. Farms with buffalo purchased from another herd were associated with IMI from any pathogen. Udder symmetry and body condition score (BCS) were risk factors related to buffalo breeding and nutrition. Asymmetrical udders were associated with IMI caused by any bacteria and with IMI caused by *S. aureus*. Poor BCS showed higher odds of IMI from any pathogen and by NASM. Progressively increasing TBC, TNAS, and TEC levels were observed along the milk value chain ($P < 0.05$). High TBC was associated with the winter season, cleaning the milk containers using pond water, a mix of buffalo and cow milk, and a poor cleanliness score for the milk containers. Farms located in coastal areas were associated with high levels of TNAS. Zoonotic pathogens' prevalence increased over the water buffalo milk value chain. Three classical enterotoxin-encoded genes for *E. coli O157:H7* were tested, *eae*, *stx1*, and *stx2*, of which the *stx2* genotype was the most prevalent in milk products (74 %). *L. monocytogenes* and *Y. enterocolitica* were more prevalent in the farm bulk milk (65-79 %) than in the later milk value chain nodes. The prevalence of *S. enterica* was low (0 - 2.9 %) in all the milk chain nodes, and all the samples were negative for *C. jejuni*. Using plastic containers and dirty clothes to clean milk containers were the risk factors associated with contamination by zoonotic pathogens in the buffalo milk value chain.

There was a high prevalence of animal and quarter level SCM, but the overall mean BMSCC was satisfactory. A higher prevalence of SCM and BMSCC was observed in intensive farms than in free-range or semi-intensive farms. Non-aureus staphylococci were the most prevalent-IMI-causing pathogen. Poor milking hygiene and cleanliness of hind quarters were associated with a higher prevalence of NASM. Overall, the prevalence of most zoonotic pathogens in milk and milk products was very high. Our findings indicate that improving udder health and hygiene practices during milk handling could help ensure the safety and quality of buffalo milk and milk products consumed in Bangladesh.